



Exploring robot service quality priorities for different levels of intimacy with service

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Received: 12 April 2023 / Accepted: 20 September 2023 / Published online: 25 October 2023
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Abstract

The influence of intimacy between humans and service robots on service quality has garnered significant attention. Applying the Mehrabian–Russell environmental psychology model, this study collected 591 valid responses from users of robots with different levels of intimacy and then used the TOPSIS technique to rank the importance of service quality. The results show, first, that high-intimacy service robots have better service quality performance. Second, for both high and low-intimacy service robots, tangibility and responsiveness are the two most important service quality dimensions. Third, the different levels of robot intimacy affect customers' ranking of the other three service quality dimensions (empathy, reliability, and assurance). This research enables scholars and professionals to comprehend the influence of robot intimacy on robot service quality, thereby facilitating successful robot service implementation.

Keywords Service robot · Intimacy · Service priorities · SERVQUAL · TOPSIS

1 Introduction

Today, AI-enabled robots can enhance the customer experience by delivering contact-free services that require no human interaction with service employees (Lee and Lee 2020). Service robots are system-based autonomous and adaptable interfaces that interact, communicate with, and deliver services to an organization's customers (Wirtz et al. 2018; Jörling et al. 2019; Huang and Rust 2021a). In situations that involve social interactions, robots should be able to establish a certain degree of automated social presence when encountering service needs; that is, they should provide consumers with a sense of personally interacting with another social entity

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(Xiao and Kumar 2021). In conjunction with the use of biometrics (such as facial and voice recognition systems), service robots can recognize individual customers and thus provide highly customized and personalized services (Wirtz et al. 2018). Currently, service robots are being developed with functions that mimic the physiological and psychological states of humans (Söderlund 2022a; Huang and Rust 2021b).

The question of how service robots influence consumers' affective feelings has attracted growing attention in academic research (Chuah and Yu 2021; Söderlund 2022b). Wirtz et al. (2018) argued that consumer acceptance of service robots depends on how well robots can deliver on functional (i.e., dominance-related) and social-emotional and relational (i.e., warmth-related) needs to achieve role congruency. Robots that can engage in affective and reactive interactions with humans are progressively developing (Chuah and Yu 2021). Robots that provide emotional support can have a positive impact on consumers' service satisfaction and other service outcomes (Becker et al. 2022a, b; Gelbrich et al. 2021; Chong et al. 2021).

These developments raise the question of whether the interactive relationship between a robot and a human will affect the quality of the service provided by the robot. Through repeated interactions, humans can develop a mental relationship with robots that is characterized by higher degrees of intimacy (Kahn et al. 2015). However, the intimacy between service robots and humans remains underexplored (Chiang and Trimi 2020); previous research has mainly focused on designing emotion recognition for service robots (Becker et al. 2022a, b; Chen et al. 2022). Intimacy, defined as openness in communicating personal thoughts, feelings, and information (Pereira et al. 2010; Leite et al. 2013), has been a core concept in research on human social exchanges (Prager 1997), being regarded as the interpersonal process of emotional communication (Reis et al. 1988), the expression of which involves both verbal and nonverbal behaviors (Prager 1997). Chiang et al. (2022) showed that anthropomorphism has a significant positive impact on intimacy between robots and humans. The investigation of how the level of intimacy between humans and service robots affects service quality is becoming increasingly important.

Furthermore, most of the previous research has focused primarily on the dimensions of service quality from an overall perspective (e.g., Choi et al. 2020; Prentice and Nguyen 2021; Zhu 2022), whereas there has been limited research ranking individual dimensions of robot service quality according to their relative importance. Chiang and Trimi (2020) pointed out that customers prioritize the importance of each service feature of service robots differently. In this study, our aim is to further explore the impact of intimacy on the ranking of robot service quality as perceived by those receiving the service. By addressing the existing gaps in understanding on the role of intimacy in service robots on service quality, we aim to contribute to the existing body of knowledge in this area.

To this end, we formulate the following research questions: (1) Does greater intimacy between service robots and customers produce better service quality performance? (2) For different intimacy levels of service robots, do customers prioritize different service quality dimensions? This study uses the Mehrabian–Russell (M–R) environmental psychology model (Mehrabian and Russell 1974) and adopts the five dimensions of service quality proposed by Parasuraman et al. (1988)—namely,

tangibility, reliability, responsiveness, assurance, and empathy. To explore the prioritization of service quality, this study examines three different intimacy service robots: a human-shaped travel consulting robot, a coffee-making robot, and an artificial intelligence speaker. The investigation collected data through a questionnaire on the three different types of service robots that provide different degrees of intimacy. Questionnaires were distributed after customers used these robot services. Data gathered from the 591 valid responses were then used to empirically test the hypotheses proposed for this study. The remainder of the paper is organized as follows. The second section presents the literature review and the hypotheses. The third section explains the research methodology. The fourth section presents the research results, and the final section discusses the findings and managerial implications for robot service delivery.

2 Literature review and hypotheses development

This study proposes that customer-perceived intimacy affects the ranking of robots' service quality dimensions, thereby implying that the perceived quality of services provided by robots is affected by the degree of intimacy that exists between the robots and humans. In this study, the M–R environmental psychology model (Mehrabian and Russell 1974) was used to formulate the hypotheses. According to that model, when humans receive stimuli from the external environment (such as visual or sensory impressions), these stimuli give rise to affective states, which in turn affect behavioral responses. This study thus sought to explore whether customers experience emotional states upon receiving external stimuli from robot services and to determine whether these states affect the customers' behavioral responses and, consequently, their assessment of the service quality received. This study also proposes that service quality affected by external emotional stimuli is intimacy-sensitive. Service quality that is not affected by emotion and emotional reactions is called basic service quality. The next section reviews the relevant literature, including the current research on service robot applications, customer-perceived intimacy of service robots, and service quality, followed by the presentation of the research hypotheses for the current study.

2.1 Service robots

As robots can provide everyday services effectively, many studies have described different service robot application cases, such as in hotels (Chiang and Trimi 2021), retail stores (Kim et al. 2022; Park and Yi 2022; Wang et al. 2022), restaurants (Acosta et al. 2006), senior care (Graf et al. 2004), and Alchatbots (Chong et al. 2021). Robots can be classified as either industrial robots or service robots based on their areas of application (Haidegger et al. 2013). Industrial robots are used in manufacturing, while service robots, which imitate humans, are used to provide automated services and can perform different tasks through different programming. According to the preliminary definition proposed by the International Federation of

Robotics (IFR 2021), service robots are robots that can operate autonomously or semi-autonomously, constitute mechanical equipment by providing useful services to humans, and perform services that are not within the scope of manufacturing (Haidegger et al. 2013; IFR IFR 2021). Based on this definition, when industrial robots (e.g., mechanical arms) perform nonmanufacturing work in nonmanufacturing industry environments, they can also be regarded as a type of service robot. For example, mechanical arms can be used in bars to serve drinks or in hospital operating rooms to assist physicians in performing surgery (Haidegger et al. 2013).

2.2 Customer-perceived intimacy of service robots

Intimacy is an important constituent element in the establishment of intimate relationships, and it has been extensively studied in the fields of sociology and social psychology (Felmlee and Sprecher 2000). McAdams (1989) defined intimacy as the willingness to share the most secret thoughts in one's heart and suggested that even though everyone has differing motives for their behavior, a craving for intimacy is one of the most basic constituent elements of human nature. Perlman and Fehr (1987) summarized intimacy in three ways: an individual's interaction with others and their mutual familiarity and dependence; the degree to which an individual is willing to reveal him- or herself; and the experience of interactions that make people feel touched and provide a sense of emotional warmth. According to Prager (1997), intimate interactions comprise intimate behaviors and intimate experiences; intimate behaviors refer to intimate sharing with a partner, which consists of the sharing of personal information and preferences, while intimate experiences refer to gaining knowledge of the partner's feelings and perceptions via the partner's behavioral responses.

One relevant related study developed a model that can effectively imitate human affective processes and then used this model to create a robotic system with a high degree of intimacy (Samani 2016). Samani (2016) further suggested that the key to effective interaction between robots and humans is the establishment of friendship, which the author defined as the gradual establishment of emotional ties through companionship. Robots detect and display different emotional states while interacting with users. They can build databases based on these interactions and keep records of past interactions with users, which can contribute to a sense of connection between the robot and its users (Huang and Philp 2021; Samani 2016). In a study using the iCat robot, Leite et al. (2013) sought to develop an intimate robot companion that could establish active social relationships with users. As people increasingly experience social interactions with anthropomorphic robots, they will establish emotional relationships with those robots that involve a high degree of intimacy and trust (Kahn et al. 2015).

The use of robots demonstrating different degrees of intimacy may have different outcomes in different service situations. Because robots should do the things at which they excel, this study examines whether affective factors can cause any changes in the ranking of robots' service quality dimensions in order of importance.

2.3 Service quality of robots

Robot service quality refers to the degree to which service results meet preset standards, and it often includes the relationship between service quality and customer satisfaction. From this perspective of the service quality dimensions of robots, some scholars have sought to identify the gap between the actual service provided by a company and the customers' expectations of each dimension (Churchill and Surprenant 1982). Service quality is a cognitive quality that is subjectively determined by customers, rather than the result of objective assessments (Garvin 1984). Certain scholars have suggested that service quality encompasses two dimensions—namely, service output (e.g., reliability) and the service process (Grönroos 1984; Levesque and McDougall 1996; Parasuraman et al. 1991). Prentice and Nguyen (2021) developed four dimensions to measure the service quality of robots: automation, personalization, efficiency, and precision. However, in Prentice and Nguyen's study, the automation dimension accounted for a high proportion of the total variance (close to 60%), which weakened the importance of the other three dimensions. Consequently, service quality reflected not only the ultimate outcome but also the method by which the service is provided, and, as a result, the quality of service outputs could be affected by the intangibility and perishability of those service outputs and by a high degree of customer participation in the service process.

Parasuraman et al. (1985) proposed a conceptual model of service quality after developing a theoretical basis through interviews with marketing, operations, upper management, and customer relationship executives, as well as interviews with consumers concerning their service impressions. They proposed that there are five gaps between the service quality expected by consumers and the actual perceived service quality, and thus it is necessary to decrease the size of these five gaps to satisfy consumers' needs fully. Parasuraman et al. (1988) modified the original ten service quality constructs into five constructs for SERVQUAL, a widely accepted and applied model to measure service quality that includes: reliability, responsiveness, assurance, empathy and tangibles (e.g., Cronin and Taylor 1992; Reynoso and Moore 1995; Sahney et al. 2004; Yeo and Li 2014). Chiang and Trimi (2020) used the modified SERVQUAL to investigate the service quality of robots, as well as to find out which service qualities are important to the service industry when robotic services are implemented. The present study also used the SERVQUAL scale to assess service quality.

3 Research hypotheses

The literature on human–robot interaction emphasizes the importance of affective sociability, including robots' ability to provide companionship (Hoffman and Vanunu 2013), increase the sociability of autistic children (Stanton et al. 2008), increase motivation (Nakagawa et al. 2011), provide affection (Nie et al. 2012), increase perceived enjoyment (Forgas-Coll 2022), and even induce ethical feelings (Kahn et al. 2012; Belk 2021). The use of robots with anthropomorphic appearances can promote affective human–robot interaction. Previous research has indicated that

when robots evoke the perception that they are similar to humans in appearance, strong senses of social presence (Kim et al. 2013) and social inclusion (Mourey et al. 2017) are induced, and people are willing to spend more time with these robots (Qiu et al. 2020).

Given the ever-increasing sociability of anthropomorphic robots, humans are increasingly forming relationships of greater perceived intimacy and trust with these robots (Kahn et al. 2015). Walters et al. (2008) discovered that people typically prefer robots with a more humanlike appearance. In an experimental study of human–machine interactions, Hinds et al. (2004) proposed that users believe that, compared to more anthropomorphic robots, robots with a more machine-like appearance tend to have relatively poorer manners and less social interactivity. Further still, users tend to believe that robots with a humanlike appearance are more functional and reliable than are robots with a mechanical appearance. Humanoid service robots make customers feel more comfortable, as these robots foster greater rapport building (Becker et al. 2022a, b). Therefore, if the robot can develop a higher degree of intimacy with its customers, then that robot's performance will be perceived to be better overall.

Soteriou and Chase (1998) used the SERVQUAL dimensions to measure the relationships between intimacy and service quality, customer waiting time, and time spent communicating with customers in the financial industry. They found that for a given waiting time, intimacy had a positive correlation with the responsiveness, concern, and tangibility perceived by the customer. Perceived empathy may thus be related to perceived intimacy. A high degree of intimacy may increase customer attention and may enhance customer awareness of the service's tangible characteristics. Regardless of the nature of the service, a high level of perceived intimacy will invariably strengthen the perception that the robotic service provider is willing to help the customer, thereby enhancing overall service quality. The current study therefore proposes the following hypothesis:

H1 A high-intimacy robot, in contrast to a low-intimacy robot, will deliver better service quality performance.

This study proposes that tangibility and responsiveness are the two basic preconditions of good robot service quality. Previous research on human–robot interaction has found that robots' presence has a significant effect on humans' interaction with and trust in those robots. Tangible services have to be performed by tangible robots that can touch, move, or process people or possessions, whereas intangible services can be provided by a range of robot types, including virtual robots such as text-based (e.g., chatbots), voice-based (e.g., Siri and Alexa), video-based, and three-dimensional virtual robots (e.g., holograph-based robots) (Wirtz et al. 2018).

For virtual robots, such as navigation devices with graphic displays, users' trust in these robots decreases when users encounter errors, regardless of whether the error originated with the user, the robot, or the user and the robot team together (Kaniarasu and Steinfeld 2014). In a study of how interaction with virtual robots and real humanoid robots affects humans' responses, Bainbridge

et al. (2008) found that while subjects would greet and interact cooperatively with both types of robots, these subjects had greater trust in the physical robots when they performed unusual tasks, implying that, for humans, robot presence influences the level of two-way social interaction. Compared to interactions with virtual robots consisting of graphic displays (Camilleri and Troise 2022; Chen et al 2021), humans' interactions with tangible robots receive more positive assessments, including greater levels of trust and respect.

Response capabilities are extremely important for all robots, since perceiving the responsiveness of others to one's needs is the basis for humans' ability to form a sense of security regarding their interaction with robots or other humans. Birnbaum et al. (2016) found that robots' response capabilities tend to enhance their appeal and make humans more willing to approach them and view them as companions when humans are in stressful situations. These scholars also discovered that, apart from having similar responses in different situations, the human interactions with responsive robots can even improve self-perception in subsequent stressful tasks. Castillo et al. (2021) found that when using chatbots, users could become discouraged and angry when experiencing poor robot responsiveness, such as when robots asked too many questions but had little understanding or when they misunderstood problems and provided irrelevant responses. Although users may not expect chatbots to completely resolve their problems, they do hope that such robots can at least understand them. Birnbaum et al. (2016) suggested that social robot responsiveness increases human nonverbal approach behaviors (physical proximity, leaning toward the robot, eye contact, smiling) and that humans can use robots as a source of consolation and security. The current study, therefore, argues that regardless of the intimacy level of robots, tangibility and responsiveness are both fundamental robot service quality indicators. Thus, the following hypothesis is offered:

H2 Regardless of whether a service robot is a high- or low-intimacy type, the two most important service quality dimensions are tangibility and responsiveness.

Previous studies have found that many people feel that the response capabilities of their companions constitute the key to the attachment process and can potentially actively influence their well-being and their relationships with those companions (Reis and Clark 2013). In particular, service robots used in the field of healthcare must show the same psychological sensitivity as the nursing staff and its concern for users' needs. For service robots with a high level of intimacy, humans may pay more attention to whether those robots display individual concern, whether the robots know what will benefit them as individuals, and whether the robots understand their individual needs. In services relying on human-robot interaction to establish deeper affective relationships, consumers often place greater emphasis on empathetic service quality, which implies that showing concern and providing personalized care are more important in those interactions. Soteriou and Chase (1998) found that for bank cashier tasks, which involve low complexity but highly differentiated service, too little intimacy may

cause customers to doubt the accuracy and reliability of the service capabilities. Subero-Navarro et al. (2022) pointed out that the affective dimension positively affects consumers' willingness to use robots in retail stores. Therefore, this study proposes that for robots with high-intimacy, the importance of empathy for service quality indicators increases when consumers attach more importance to the robots' empathy.

Chiang et al. (2022) suggested that a high degree of intimacy tends to make humans think that robots are knowledgeable and courteous, which can foster trust in those robots. Thus, robots have the quality of assurance, which refers to the ability to perform tasks with expertise, politeness, and trust. Such close relationships also enable the robots to provide accurate services that meet or exceed customers' needs and preferences. Thus, robots have the quality of reliability, as they appear to reliably perform the committed or requested services. Therefore, this study proposes that empathy, assurance, and the reliability of service quality can be affected by intimacy and uses these three indicators as intimacy-sensitive indicators. Of these, empathy is the most important service quality to have in high-intimacy services. Therefore, this study proposes the following hypotheses:

H3a For high-intimacy robot services, the most important service quality dimensions, in order, are tangibility, responsiveness, empathy, assurance, and reliability.

H3b For low-intimacy robot services, the most important service quality dimensions, in order, are tangibility, responsiveness, assurance, reliability, and empathy.

4 Methodology

4.1 Subjects

This study employed a questionnaire survey to collect data from consumers who use robot services on site. The study included three types of subjects: Users of Zenbo, users of a Aibot Caffeine robot, and users of a Mi AI speaker. The first of these, Zenbo, is a travel agency robot, and its abilities include consulting and making calls. The Zenbo robot has a humanlike appearance and form. It can provide responses and change its facial expressions upon receiving commands from humans; for example, when a human strokes Zenbo's head, the robot can display a shy but pleased expression. In addition, people can use the Zenbo robot to query for information. For example, customers can ask Zenbo to search for a good restaurant with local cuisine and make a reservation. Next, the Aibot is a coffee-making robot that uses mechanical arms to brew coffee. The Aibot can use its arms to press the button of a syrup bottle to add sugar, and it can place a lid on a cup of brewed coffee and hand it to a consumer. Finally, the Mi AI speaker is a rectangular upright smart speaker that can answer human questions or play music. For example, it can answer questions like "What will the weather be today?" and "How long does it take to get from home to the conference room?" Data collection for Zenbo and the Mi AI speaker was done in cooperation with a travel agency in Taipei City. For this study, Zenbo and the Mi

AI speaker were placed in the travel agency, allowing customers to use these two types of robots in the travel agency. Meanwhile, Aibot is used to sell coffee in a department store. The current study's questionnaire survey was conducted after the subjects used the Zenbo and Mi AI speaker robots in the travel agency or purchased coffee made by the Aibot at the department store. This study used samples collected from three different robotic services for the analysis because different kinds of services can create different degrees of intimacy.

4.2 Data collection

This study collected questionnaires from August 2018 through November 2019. A total of 300 questionnaires on the Zenbo robot's service were distributed, and 207 valid questionnaires were returned. A total of 300 questionnaires regarding the Aibot robot's service were also distributed, and 202 valid questionnaires were returned. Finally, 300 questionnaires regarding the Mi AI speaker were distributed, and 182 valid questionnaires were returned. Thus, 591 valid responses were returned, for a valid response rate of 65.67%. The SPSS 23 software was used to perform a demographic statistical analysis (see Table 1). Of the respondents, 64.5% were women, and 35.5% were men. Individuals between 25 and 39 years in age accounted for 24.4% of all the respondents. For level of education, junior college or university graduates accounted for the largest percentage (51.6%) of the respondents.

Table 1 Demographics of the respondents

Variable	Category	Frequency	Percentage (%)
Gender	Male	381	64.5
	Female	210	35.5
Age	18–24	127	21.5
	25–39	144	24.4
	40–54	101	17.1
	55–69	133	22.5
	70–84	78	13.2
	85 ore more	8	1.4
Education	Elementary school or below	49	12.4
	Middle school	51	8.6
	High school	107	18.1
	Junior college or university	305	51.6
	Graduate school and above	55	9.3
Previous experience with receiving robot services at other places (number of times)	Never	378	64.0
	1	104	17.6
	2	55	9.3
	3	11	1.9
	4 or more	43	7.3

The total sample size was 591

To confirm whether there was a significant difference between valid and invalid samples, we performed a nonresponse bias test. In addition, because it took more than a year to collect the data, we also tested for any nonresponse bias between the data collected in the first six months and the data collected in the last six months. Both results showed *p*-values for chi-square tests that were greater than 0.05, indicating no significant difference between valid and invalid samples and no significant difference between the different times of data collection.

4.3 Measures

The intimacy dimension for this study was modified from the definitions of Leite et al. (2013), Pereira et al. (2010), and Kahn et al. (2015). The operational definition of intimacy is customers' perception of robots' sensitivity to customers' needs and their affective state. The questionnaire items concerning service quality dimensions were based on the service quality literature (e.g., Parasuraman et al. 1985, 1988; Cronin and Taylor 1992; Chiang and Trimi 2020), and the five major dimensions of service quality—namely, tangibility, reliability, responsiveness, assurance, and empathy. The operational definitions of these dimensions of robot service quality are as follows: tangibility refers to robots' modern physical equipment and professional appearance; reliability refers to robots' ability to reliably perform the promised service; responsiveness refers to robots' willingness to help customers and provide prompt service; assurance refers to robots' knowledgeability and ability to induce customer trust; and empathy refers to the concern and personalized care robots show for customers. Table 2 presents the operational definitions of the dimensions used in this study. A 7-point Likert scale was used for service quality, while a 5-point Likert scale was used for intimacy.

Since most of the constructs utilized in the questionnaire were adopted from previous studies, the questionnaire development process utilized the back-translation method (Brislin 1980). First, the questionnaire was developed in English by the authors. Then, it was translated into Chinese by two bilingual professors in the field of service management. Finally, the Chinese version of the questionnaire was translated back into English by two other bilingual professors specializing in service management. The two English versions of the questionnaire had no significant differences and required only minor adjustments.

4.4 Technique for order preference by gauging the similarity to the ideal solution (TOPSIS)

The TOPSIS ranking method, first proposed by Hwang and Yoon (1981), is used to determine the ranking order of the strengths and weaknesses obtained from data, determine a solution's degree of proximity to the optimal solution, and use that solution as a basis for assessing the subject's strengths and weaknesses. This study used the TOPSIS method to investigate differences in consumers' ranking of service robot performance and employed the R language to write the TOPSIS programming code to rank the dimensions of service quality.

Table 2 Operational definitions of research constructs

Dimension	Operational definition	Source
Intimacy	The robot communicates its own thoughts, feelings, and information to customers	Leite et al. (2013), Pereira et al. (2010)
Reliability	The robot can reliably perform its promised services	Parasuraman et al. (1985, 1988, 1991), Chiang and Trimi (2020)
Responsiveness	The robot is willing to help the customer and provide prompt service	Parasuraman et al. (1985, 1988, 1991), Chiang and Trimi (2020)
Assurance	The robot is knowledgeable and can inspire customers' trust	Parasuraman et al. (1985, 1988, 1991), Chiang and Trimi (2020)
Empathy	The robot exhibits concern for the customer and personalized care	Parasuraman et al. (1985, 1988, 1991), Chiang and Trimi (2020)
Tangibility	The robot has modern physical equipment and a professional appearance	Parasuraman et al. (1985, 1988, 1991), Chiang and Trimi (2020)

4.5 Common method variance

Because this study sought to determine the existence of common method variance (CMV), CMV testing was performed on the questionnaire items for all the dimensions. To ensure that a certain common factor did not explain the most variance, this study used Harman's single-factor test in the analysis. The study also used the SPSS software to perform the explorative factor analysis (EFA) of all items and determine the existence of CMV in the questionnaire. The total maximum variance in the first component of the unrotated variables was 9.255, and the variance in the explanatory ability of all variables was less than 50% (44.073%); therefore, severe CMV was not present in this study.

A CFA comparison was used to further confirm that CMV was not an issue. After Model 1 was established with the AMOS software (single-factor model), all items were tested by placing them into the same CMV dimension. Model 1 had a goodness-of-fit index (GFI) of 0.625, an adjusted goodness-of-fit index (AGFI) of 0.585, a comparative fit index (CFI) of 0.643, and a root mean square error of approximation (RMSEA) of 0.148, indicating a relatively poor fit. Model 2 (multifactor model) placed the items into the corresponding dimensions in accordance with the literature, and each CMV dimension then was assumed to have an identical effect on the observed variables. The analysis results revealed that Model 2 had a GFI of 0.898, AGFI of 0.863, CFI of 0.928, and RMSEA of 0.073. The chi-squared values for Model 1 and Model 2 can be used to confirm whether there was a CMV issue. These results show that the difference in the chi-squared values between the single-factor model and the multifactor model was significant, further confirming that severe CMV did not exist. Because severe CMV was not present, we have confidence in the relationship established by the statistical results.

5 Results

5.1 Reliability and validity analysis

The CFA of each dimension yielded a normed fit index (NFI) of 0.908, a CFI of 0.928, and a RMSEA of 0.072, indicating that the CFA model overall had an excellent fit. The higher the value of Cronbach's α is, the greater is the level of credibility. The Cronbach's α was > 0.7 , indicating acceptable results (Cortina 1993). This study consequently used Cronbach's α to determine whether the questionnaire had internal consistency, and reliability was determined for each of the dimensions—reliability, responsiveness, assurance, empathy, tangibility, and intimacy (see Table 3). The Cronbach's α values for all the dimensions were greater than 0.7, signifying that each dimension had excellent internal consistency and that the measurement instrument both was appropriate and had good stability.

The composite reliability (CR) of all the variables was tested, with the goal of determining the internal consistency of the dimensions. The CR calculation formula and the standardized factor loading of individual variables are related to the variables' measurement error. The higher the CR value is, the greater is

Table 3 Construct measurements with factor loadings

Construct	Items	Factor loadings	Cronbach's α	CR	AVE
Intimacy	I perceived that robot showed sensibility toward my affective state	0.750	0.868	0.872	0.695
	I perceived that robot knew when something was bothering me	0.845			
	I perceived that robot knew when I was upset	0.899			
Reliability	The robot can provide services as promised	0.634	0.855	0.859	0.550
	The robot dependably handles customers' service problems	0.757			
	The robot performs services right the first time	0.797			
	The robot provides services at the promised time	0.791			
	The robot maintains error-free records	0.717			
Responsive-ness	The robot keeps customers informed about when services will be performed	0.693	0.837	0.840	0.569
	The robot provides prompt service to customers	0.767			
	The robot shows willingness to help customers	0.754			
	The robot shows readiness to respond to customers' requests	0.799			
	The robot instills confidence in customers	0.929			
Assurance	The robot makes customers feel safe in their transactions	0.904	0.913	0.913	0.840
	The robot has the customer's best interest at heart	0.779			
Empathy	The robot understands the needs of customers	0.823	0.772	0.783	0.550
	The robot has convenient business hours	0.605			
	The robot has modern equipment	0.711			
	The robot has visually appealing facilities	0.736			
	The robot has a neat, professional appearance	0.845			
Tangibility	The robot has visually appealing materials associated with the service	0.786	0.849	0.854	0.595

Model fit: $\chi^2/df = 4.098$, $NFI = 0.908$, $CFI = 0.928$, $IFI = 0.929$, $RMSEA = 0.072$, $SRMR = 0.052$

All factor loadings are the completely standardized solution

the internal consistency of the dimensions. As suggested by Fornell and Larcker (1981), CR values should be greater than 0.6. In this study, each dimension had a CR value greater than 0.6, indicating that all the dimensions in this study had good internal consistency (see Table 3).

To determine validity, both convergent validity and discriminant validity were analyzed. Convergent validity refers to the degree of correlation between dimensions that theoretically should be related (Campbell and Fiske 1959). Fornell and Larcker (1981) proposed that the standard for evaluating convergent validity is that the standardized factor loading of all items must be greater than 0.5, the *t*-value must be significant, and the average variance extracted (AVE) should be greater than 0.5. The convergent validity analysis results for all dimensions in this study indicated that the AVE values for all dimensions were greater than the standard value of 0.5 and all standardized factor loading values were significant, signifying excellent convergent validity for the measurement instrument (see Table 3).

The purpose of discriminant validity is to understand whether there are differences in the correlation analysis between two designated dimensions. To avoid measuring the same item between dimensions, a discriminant validity test was conducted. According to Fornell and Larcker (1981), the correlation coefficient between dimensions should be less than the square root of the AVE value for each dimension; that is, when the square root of the AVE value is greater than the correlation coefficient between dimensions, then the scale has good discriminant validity. In this study, the square root of the AVE value for each dimension was greater than all dimension's correlation coefficients, signifying excellent discriminant validity for each variable (see Table 4).

Table 4 Descriptive statistics and correlations

Construct	Mean (SD)	1	2	3	4	5	6
1. Reliability	4.773 (1.114)	0.742/0.859					
2. Responsiveness	5.085 (1.038)	0.708**	0.754/0.840				
3. Assurance	4.991 (1.250)	0.669**	0.716**	0.917/ 0.913			
4. Empathy	4.723 (1.192)	0.537**	0.600**	0.550**	0.742/0.783		
5. Tangibility	5.581 (0.996)	0.526**	0.596**	0.551**	0.608**	0.771/0.854	
6. Intimacy	2.742 (0.955)	0.350**	0.324**	0.291**	0.505**	0.344**	0.834/0.872

The values below the diagonal are correlations. The diagonal values in italics to the left of the slash are the square root of the average variance extracted (AVE) for each construct; the values to the right are the composite reliabilities

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

5.2 Independent samples t-test

This study used the independent samples t-test to test whether a significant difference in service quality existed between the high-intimacy and low-intimacy groups. The results indicated that the reliability, responsiveness, assurance, empathy, and tangibility were significantly different between the high- and low-intimacy groups ($p < 0.000$) and also that the service quality in the high-intimacy group was uniformly higher than it was in the low-intimacy group, confirming Hypothesis 1 (see Table 5). Thus, a high-intimacy robot, in contrast to a low-intimacy robot, will deliver better service quality performance.

5.3 TOPSIS ranking order of service outcomes

In this study, TOPSIS was used to analyze the ranking of service qualities. First, the data were sorted. Second, the customers' perceived intimacy was calculated, and they were divided into high- and low-intimacy groups. Next, a TOPSIS ranking analysis was conducted to compare the influence of high-intimacy and low-intimacy on service quality dimension rankings. These ranking results are shown in Table 6. The ranking order for all the data were: tangibility, responsiveness, assurance, reliability, and empathy. The ranking order for the high-intimacy group was: tangibility, responsiveness, empathy, assurance, and reliability. The ranking order for the low-intimacy group was: tangibility, responsiveness, assurance, reliability, and empathy. These results indicate that tangibility and responsiveness constituted a basic service quality, while empathy, assurance, and reliability constituted an intimacy-sensitive service quality, which changes as robot intimacy varies. This finding confirms Hypotheses 1, 3a, and 3b. Thus, regardless of whether a service robot is a high- or low-intimacy type, the two most important service quality dimensions are tangibility and responsiveness. In addition, in services where the human and the robot can

Table 5 T-test of independent samples

Construct	Mean and SD	Overall dataset ($n = 591$)	High intimacy robots ($n = 289$)	Low intimacy robots ($n = 302$)	<i>t</i> -value
Reliability	Mean	4.773	5.098	4.462	7.234***
	SD	1.114	1.035	1.099	
Responsiveness	Mean	5.085	5.320	4.861	5.507***
	SD	1.038	0.946	1.073	
Assurance	Mean	4.991	5.280	4.714	5.651***
	SD	1.250	1.173	1.261	
Empathy	Mean	4.723	5.192	4.275	10.115***
	SD	1.192	1.081	1.120	
Tangibility	Mean	5.581	5.811	5.361	5.655***
	SD	0.996	0.905	1.030	

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6 TOPSIS ranking

Overall dataset (<i>n</i> = 591)	Ranking of high-intimacy robots (<i>n</i> = 289)	Ranking of low-intimacy robots (<i>n</i> = 302)
Tangibility (0.751)	Tangibility (0.747)	Tangibility (0.753)
Responsiveness (0.534)	Responsiveness (0.521)	Responsiveness (0.540)
Assurance (0.465)	Empathy (0.449)	Assurance (0.477)
Reliability (0.408)	Assurance (0.442)	Reliability (0.412)
Empathy (0.385)	Reliability (0.400)	Empathy (0.350)

The numbers in parentheses are sorting weights

establish an affective relationship, consumers place a premium on empathetic service quality.

6 Discussion

6.1 Conclusion

This study employed the M–R environmental psychology model to develop an empirical model and used empirical data to investigate whether robot service quality is affected by the level of intimacy between robots and customers. The key findings are as follows.

First, high-intimacy tends to enhance robot service quality. In other words, the deeper is the emotional relationship between robots and humans that is established through communication and interaction, the better is the robot service quality. This study found that after people received robot services, affective states were generated to different degrees in accordance with their interactions, and these generated states further influenced the individuals' approach and avoidance behavioral responses and thus their overall assessment of robot service quality.

Second, tangibility and responsiveness are the two most important service quality dimensions for achieving customer satisfaction. In this study, tangibility refers to robots' modern physical equipment and professional appearance, while responsiveness refers to robots' willingness to help customers and provide prompt service. Regardless of whether the services are provided by robots or real humans, these two service quality dimensions give customers a sense of the robot's proper deportment and response capabilities, both of which are prerequisites for customer satisfaction.

Third, in services where the human–robot interaction provides a basis for the establishment of an affective relationship, consumers place a premium on empathetic service quality; that is, providing showing concern and personal attention toward customers is relatively important in those cases. Conversely, for low-intimacy services, consumers place little emphasis on empathetic services, a finding that is consistent with intuition. Interestingly, however, regardless of the intimacy level of the services, the assurance and reliability of services both ranked quite low. This result may be attributable to the still-limited state of robotic technology.

Because robotic technology still has considerable room for improvement, and AI is not-yet-mature, robots cannot necessarily perform their tasks reliably in many cases, a factor that makes customers hesitant to trust the services they provide.

6.2 Theoretical implications

This study provides several theoretical implications. First, it contributes to the service robot literature by demonstrating that robot intimacy affects robot service quality performance. The topic of robot emotional intelligence is gaining attention in academic research (Rust and Huang 2021; Vitezić and Perić, 2021). Service robots that provide emotional support have been shown to have a positive impact on consumer satisfaction with services (Gelbrich et al. 2021). Most of the previous empirical research has focused mainly on designing emotion recognition for service robots (Becker et al. 2022a, b; Chen et al. 2022), while little research has been done on whether overall service quality can be improved by optimizing robot service. Chiang and Trimi (2020) pointed out that the dimensions of robot service quality valued by individual customers vary based on their emotions. Consistent with previous research, we find that more intimate service robots make customers perceive better service quality.

Second, this study further contributes to the service robot literature by exploring the priority of service quality at different levels of robot intimacy. Most previous service robot research has examined service quality from an overall perspective (e.g., Choi, et al. 2020; Shimmura et al. 2020; Zhang et al. 2022) but has not ranked individual service quality dimensions according to different intimacy levels of service robots. This study contributes the finding that customers prioritize different service quality dimensions for different intimacy levels of service robots. This study suggests that the ranking order for the high-intimacy robots was tangibility, responsiveness, empathy, assurance, and reliability, while the ranking order for the low-intimacy robots was tangibility, responsiveness, assurance, reliability, and empathy. This suggests that tangibility and responsiveness constituted a basic service quality. Previous research has also found that higher levels of trust and respect results from human interaction with tangible and responsive robots (Birnbaum et al. 2016; Camilleri and Troise 2022; Chen et al 2021).

Third, this study contributes the finding that empathy is a service quality aspect that customers value more in high-intimacy services. This study proposes that empathy, assurance, and reliability of service quality can be affected by intimacy and uses these three indicators as intimacy-sensitive indicators. This finding is consistent with previous studies (Tian and Oviatt 2021; Tojib et al. 2023) that demonstrated the benefits of artificial empathy in enabling a robot to interact in a more socially acceptable way. This study showed that such artificial empathy can fulfill customers' need for affective support, which is necessary for a more satisfying customer service experience. Researchers may recognize the variations in service quality dimensions based on the varying levels of intimacy displayed by service robots.

The results of this study are consistent with the human–robot interaction literature, which emphasizes the importance of affective sociability, i.e., the ability of

robots to provide companionship (Hoffman and Vanunu 2013), increase motivation (Nakagawa et al. 2011; Song and Kim 2022), provide affection (Nie et al. 2012), perceived enjoyment (Forgas-Coll, 2022), and even induce ethical feelings (Kahn et al. 2012; Belk 2021). This research extends the literatures (Apraiz et al. 2023) by demonstrating the reliance on human–robot interaction to establish deeper affective relationships in services. Consumers place a higher value on empathetic service quality, which means that concern and personalized attention are more important in these interactions.

This research contributes to bridging the SERVQUAL dimensions with human–robot interaction by investigating the correlations between intimacy and robot service quality. Service quality is a cognitive quality that is subjectively determined by customers, rather than the result of objective assessments (Garvin 1984). From this perspective of robot service quality dimensions, perceived empathy may thus be related to perceived intimacy. In the case of low-intimacy services, consumers tend to place less emphasis on empathetic services, which aligns with common intuition. These findings have important implications for the human–robot interaction literature.

6.3 Practical implications

The study's findings have several implications for management. First, the software engineers who are involved in robot design should incorporate a degree of intimacy appropriate to the services provided by those robots. If robot services have an emotional component, then robots should be as humanlike as possible and capable of in-depth interactions and communication with humans; the establishment of such affective relationships can enhance service quality. However, when robots provide task-oriented services, they need only to be designed with a certain degree of humanlike qualities; there is no need for them to engage in in-depth affective interactions with humans to achieve good service quality. This finding can provide useful guidance to service firms that wish to adopt robotic services, and it is consistent with the suggestion made by DiSalvo et al. (2002) that robot morphology should be based on the robot's intended function. When a robot is designed to perform human tasks, its form should convey a degree of "product characteristics" that can help users feel comfortable using it. Similarly, if companionship is important, a robot should project a degree of "humanity" that can make users feel comfortable when humans interact socially with the robot (Fong et al. 2003). However, robot design should also incorporate a degree of "roboticity," which is a necessary feature to ensure that users do not develop false expectations of the robot's capabilities (Duffy 2003). Furthermore, when robots are employed to provide services, the foremost requirement of such services is that they be tangible and responsive. In addition, the application of AI in robot services should be accelerated, and the reliability and assurance of service quality should be strengthened to improve customers' overall experience.

Companies should also emphasize flexibility in their competitive priorities to match the service quality dimensions valued by customers. The current study suggests that when robots are initially used to provide services, an enterprise should

employ a flexible approach to compensate for possibly unreliable service and a lack of trust resulting from using not-yet-mature AI. In particular, for high-intimacy services, flexibility within the enterprise can enhance empathy when delivering service quality. To ensure that consumer dissatisfaction with services does not increase, enterprises must be ready to change their competitive priorities whenever robots are initially used to provide services.

6.4 Limitations and future research

This study has the following limitations. First, the data were collected before the COVID-19 pandemic. As the study has noted the growing trend of retail stores implementing service robots, it can be considered as a preliminary study. Further research can be conducted in the post-pandemic era to examine a larger sample of retail stores that have implemented service robots. Comparing the effects of COVID-19 to the current research model may be a valuable aspect of future studies. Second, while the adoption of robots for service processes provides new business models for service industries, the use of robots is not yet widespread, the development of robot software and technologies is still-limited, and the literature rarely discusses the application of robots for service management. As a result, this study has only a limited basis of theory and literature upon which to draw. Third, owing to time, space, and funding constraints, this study used only three types of robots as its research targets: the Zenbo household companion robot, the Aibot Caffeine coffee-making robot, and the Mi AI speaker. The research samples consequently did not cover a broad spectrum of robot service applications, nor did these samples encompass the use of robot services by respondents in different geographical areas. Future research should investigate more types of robots in different servicescapes. Fourth, this study gathered samples from practical service sectors, including department stores and travel agencies. Other factors may also affect the results, such as personnel and the business environment. Further studies could use experimental designs to investigate consumer behavior and provide in-depth analysis. Nonetheless, the present results help scholars and practitioners understand how service quality could be affected if a high-intimacy robot is introduced into service processes. The results of this study thus may serve as a reference for companies in the service industry as they optimize their robotic service in the future.

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